Statistics 2 2026/27 Exam - Solutions

Introduction

The dataset `amsterdam-house-prices.csv` contains information on the asking price of a random sample of houses for sale in Amsterdam in August 2021, as well as some basic information on the listings.

The definitions of all the variables are as follows:

- `price`: The asking price in thousands of euros.
- `area`: The floor space in square meters (m^2).
- `bedrooms`: The number of bedrooms.
- `neighborhood`: Whether the listing was in the North, South, East, West, or Central neighborhood of Amsterdam.

Use this dataset to answer the following questions.

Question 1

What is the sample correlation between `price` and `area`?

Answer:

```
df <- read.csv("amsterdam-house-prices.csv")
cor(df$price, df$area)
```

```
[1] 0.7352108
```

Model 1

Estimate a linear regression model with `price` as the dependent variable and `area` as the independent variable.

If you estimated the model correctly, you should have an estimated intercept of 23.3385.

Answer:

```
m1 <- lm(price ~ area, data = df)
summary(m1)
```

```
Call:
lm(formula = price ~ area, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-547.71 -132.44   17.14   93.88  974.07

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  23.3385    17.2882    1.35    0.177
area          5.9623     0.1876   31.79   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 190.1 on 859 degrees of freedom
Multiple R-squared:  0.5405,    Adjusted R-squared:   0.54
F-statistic:  1011 on 1 and 859 DF,  p-value: < 2.2e-16
```

Question 2

What is the sample regression slope?

Answer:

```
coef(m1)[2]
```

```
    area
5.962338
```

The slope is 5.9623.

Question 3

According to the sample regression slope, a 1m^2 increase in the area of a house on average increases the listing price by how many euros?

Note: Pay attention to the units of the variables. Price is measured in *thousands* of euros. You will need to multiply by 1000 to report your answer in euros. Pay attention to the units in all questions that follow.

Answer:

The question is referring to a one-unit increase in `area`. However, price is measured in thousands of euros. Therefore we need to multiply the estimated slope by 1000 to get the answer:

```
coef(m1)[2] * 1000
```

```
    area
5962.338
```

Question 4

Plot the residuals (on the vertical axis) against `area` (on the horizontal axis). Choose the answer below which best describes what you see:

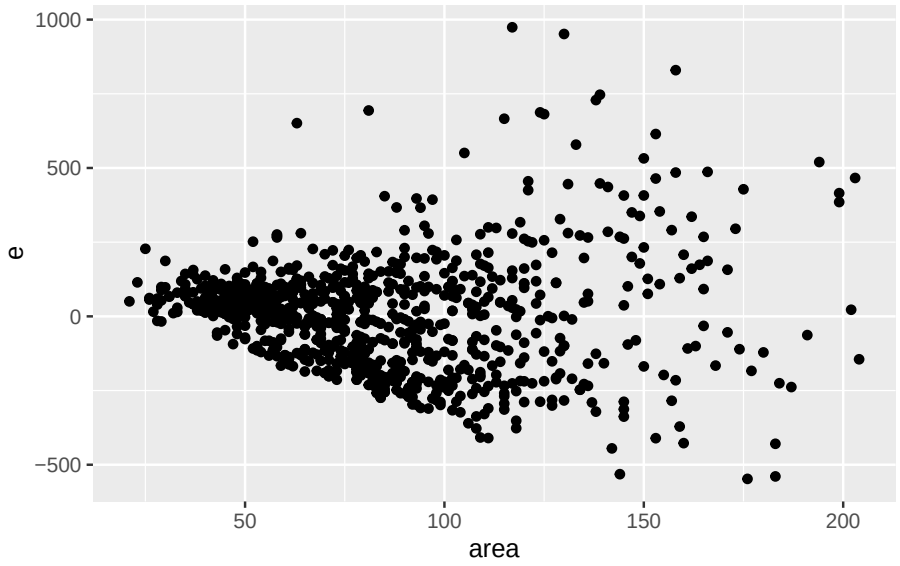
- The dispersion of the residuals increases as area increases.
- The dispersion of the residuals decreases as area increases.
- The dispersion of the residuals is constant in area.
- The residuals are all positive.
- The residuals are all negative.

Based on this, which model assumption is likely violated?

- Linear in parameters
- Random sampling
- Sample variation in the explanatory variable
- Zero conditional mean
- Homoskedasticity
- Normality

Answer:

```
df$e <- m1$residuals
library(ggplot2)
ggplot(df, aes(area, e)) + geom_point()
```



The dispersion of the residuals increases as area increases. Therefore the homoskedasticity assumption is likely violated.

Model 2

Estimate a linear regression model with `price` as the dependent variable and `bedrooms` as the independent variable.

If you estimated the model correctly, you should have an estimated intercept of 166.604.

```
m2 <- lm(price ~ bedrooms, data = df)
summary(m2)
```

```
Call:
lm(formula = price ~ bedrooms, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-479.62 -152.61  -48.11   76.89 1251.89

Coefficients:
            Estimate Std. Error t value       Pr(>|t|)
(Intercept)  166.604     26.754   6.227 0.000000000742 ***
bedrooms     110.503      7.646  14.452        < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 251.5 on 859 degrees of freedom
Multiple R-squared:  0.1956,    Adjusted R-squared:  0.1947
F-statistic: 208.9 on 1 and 859 DF,  p-value: < 2.2e-16
```

Question 5

The estimated intercept of 166.604 can be interpreted as follows.

The expected price of houses with _____ bedrooms is €_____. However, this estimate is unreliable because:

- This number of bedrooms is outside the range of our data.
- The estimated intercept is not statistically different from zero.
- The estimated slope is not statistically different from zero.
- The adjusted R^2 is smaller than the standard R^2.

Answer: 0; 166604; This number of bedrooms is outside the range of our data.

Question 6

Perform a formal test for heteroskedasticity by regressing the square of the residuals on `bedrooms`.

Use a 5% significance level.

In this test, the null hypothesis is that the model exhibits which of the following:

- Homoskedasticity
- Heteroskedasticity
- Multiskedasticity
- Paraskedasticity

What is the p-value from this test? _____

What is the conclusion? Choose one of the answers below:

- The model exhibits heteroskedasticity. Therefore the standard errors are not reliable.
- The model exhibits homoskedasticity. Therefore the standard errors are not reliable.
- The model exhibits heteroskedasticity. Therefore the standard errors are reliable.
- The model exhibits homoskedasticity. Therefore the standard errors are reliable.

Answer:

```
df$e2 <- m2$residuals^2
summary(lm(e2 ~ bedrooms, data = df))
```

```
Call:
lm(formula = e2 ~ bedrooms, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-134063  -51975  -27160   -3069 1512554

Coefficients:
            Estimate Std. Error t value      Pr(>|t|)
(Intercept)   -25605      15290  -1.675        0.0944 .
bedrooms       26758       4370   6.123 0.00000000139 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 143700 on 859 degrees of freedom
Multiple R-squared:  0.04183,   Adjusted R-squared:  0.04071
F-statistic:  37.5 on 1 and 859 DF,  p-value: 0.000000001392
```

Under the null hypothesis the model exhibits homoskedasticity. The p-value from the test is 0.000000001392. The conclusion is: *The model exhibits heteroskedasticity. Therefore the standard errors are not reliable.*

Model 3

Estimate a regression model with `price` as the dependent variable and the following 2 independent variables:

1. `area`
2. `bedrooms`

If you estimated the model correctly, your estimated intercept should equal 92.531.

Answer:

```
m3 <- lm(price ~ area + bedrooms, data = df)
summary(m3)
```

```

Call:
lm(formula = price ~ area + bedrooms, data = df)

Residuals:
    Min      1Q  Median      3Q     Max 
-574.74 -124.03    4.55   90.71  915.84 

Coefficients:
            Estimate Std. Error t value       Pr(>|t|)    
(Intercept)   92.531     19.940   4.640 0.000004021021 ***
area           7.265      0.271  26.807        < 2e-16 ***
bedrooms     -54.465      8.350  -6.523 0.000000000118 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 185.6 on 858 degrees of freedom
Multiple R-squared:  0.5622,    Adjusted R-squared:  0.5612 
F-statistic:   551 on 2 and 858 DF,  p-value: < 2.2e-16
```

Question 7

What proportion of the variation in `price` is left unexplained by the model?

Note: your answer should be between 0 and 1.

Answer:

The R-squared measures the proportion of the variation in the `price` data that is explained by our model. The proportion left unexplained is one minus this. We can calculate this as follows:

```
1 - summary(m3)$r.squared
```

```
[1] 0.4377576
```

Question 8

Use the model to test the following claim at the 5% level:

"An additional bedroom in a listing on average decreases the asking price by more than €40,000, holding the area fixed."

Perform this test by answering the questions below. Use the conventional standard errors given by `summary()`.

- What is the null hypothesis? $\beta_2 < / \leq / > / \geq / = / \neq$ _____ (choose one comparison operator and fill in a value in the blank).
- What is the alternative hypothesis? $\beta_2 < / \leq / > / \geq / = / \neq$ _____ (choose one comparison operator and fill in a value in the blank).
- Under the null hypothesis, the test statistic $T = (B_2 - b)/S_{B_2}$, where b is the hinge, follows a t distribution with how many degrees of freedom? _____
- What is the value of the test statistic? _____
- What is the associated p-value? _____
- What is your conclusion? Choose an option below:
  - Reject H0: There is sufficient evidence for the claim.
  - Reject H0: There is not sufficient evidence for the claim.
  - Don't reject H0: There is sufficient evidence for the claim.
  - Don't reject H0: There is not sufficient evidence for the claim.

Answer:

The interpretation of the slope coefficient is as follows: on average, one additional bedroom decreases the asking price of a listing by €54,465, holding the area fixed. Thus, we estimate that it decreases by more than €40,000. The claim in the question is asking if this estimate from our sample is precise enough to say the effect is bigger than €40,000 in the population. Thus the claim is that $\beta_2 < -40$. This is the alternative hypothesis. So we get:

- $H_0 : \beta_2 \geq -40$.
- $H_1 : \beta_2 < -40$.

The number of degrees of freedom is $n-k-1$, which is 858 (can be read from the summary output). The value of the test statistic can be calculated as follows:

```
b_2 <- coef(summary(m3))["bedrooms", "Estimate"]
s_b_2 <- coef(summary(m3))["bedrooms", "Std. Error"]
t <- (b_2 + 40) / s_b_2
t
```

```
[1] -1.732301
```

The associated p-value from this lower tail test is:

```
pt(t, 858)
```

```
[1] 0.04178966
```

This is smaller than 0.05, so we reject H_0. Therefore the answer is *Reject H0: There is sufficient evidence for the claim.*

Question 9

Use the estimated model to obtain a 95% confidence interval for the average price of 3-bedroom listings that are 80m^2 in size. Report your answer in euros.

Answer:

```
df_p <- data.frame(area = 80, bedrooms = 3)
predict(m3, df_p, interval = "confidence", level = 0.95)
```

```
       fit      lwr     upr
1 510.3518 497.4246 523.279
```

Lower bound: €497,424.60

Upper bound: €523,279.00

Model 4

Estimate a regression model with `price` as the dependent variable and the following 4 independent variables:

1. A dummy variable that equals 1 if the listing is in the East and 0 otherwise.
2. A dummy variable that equals 1 if the listing is in the North and 0 otherwise.
3. A dummy variable that equals 1 if the listing is in the South and 0 otherwise.
4. A dummy variable that equals 1 if the listing is in the West and 0 otherwise.

Note: By using the variable `neighborhood` as is in your model formula, these dummy variables will be created automatically.

If you estimated the model correctly, your estimated intercept should equal 719.41.

Answer:

```
m4 <- lm(price ~ neighborhood, data = df)
summary(m4)
```

```
Call:
lm(formula = price ~ neighborhood, data = df)

Residuals:
```

```
    Min      1Q  Median      3Q     Max
-479.41 -169.41  -69.41   99.71 1344.71

Coefficients:
                  Estimate Std. Error t value         Pr(>|t|)
(Intercept)         719.41      21.87  32.892          < 2e-
16 ***
neighborhoodEast   -269.12      28.64  -9.398          < 2e-
16 ***
neighborhoodNorth  -246.70      34.29  -7.194 0.00000000000137 ***
neighborhoodSouth   -39.11      33.06  -1.183            0.237
neighborhoodWest   -249.68      26.22  -9.521          < 2e-
16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 258.8 on 856 degrees of freedom
Multiple R-squared:  0.1511,    Adjusted R-squared:  0.1471
F-statistic: 38.08 on 4 and 856 DF,  p-value: < 2.2e-16
```

Question 10

According to the estimated model, what is the average price of listings in the central neighborhood?

Note: Report your answer in euros, taking note of the units of your variables.

Answer:

This is just the intercept multiplied by 1,000, as the central neighborhood is the baseline category. So the answer is €719,410.

Question 11

Let $\bar{p}_j$ be the average price (in €1,000s) of a listing in neighborhood j, where j is one of Central, East, North, South or West.

The estimated coefficient in front of the East neighborhood dummy in the model is -269.12. Which of the following does this number represent?

- $\bar{p}_{East}$
- $\bar{p}_{East} - \bar{p}_{Central}$
- $\bar{p}_{Central} - \bar{p}_{East}$
- $\bar{p}_{Central} + \bar{p}_{East}$
- $\bar{p}_{Central}$

Answer:

Let j be the neighborhood. The model is:

$$\mathbb{E}[Y|j] = \beta_0 + \beta_1 \mathbf{1}\{j = East\} + \beta_2 \mathbf{1}\{j = North\} + \beta_3 \mathbf{1}\{j = South\} + \beta_4 \mathbf{1}\{j = West\}$$

According to the model:

$$\mathbb{E}[p_i|j = East] = \beta_0 + \beta_1$$

And:

$$\mathbb{E}[p_i|j = Central] = \beta_0$$

The estimate β_1 is therefore an estimate of $\mathbb{E}[p_i|j = East] - \mathbb{E}[p_i|j = Central]$, which is $\bar{p}_{East} - \bar{p}_{Central}$.

Question 12

Test the following claim at the 5% level: "There are differences in average house prices across neighborhoods".

- The null hypothesis is that *all/at least one/none* of $\beta_j =$ _____ for j from _____ to _____.
- The alternative hypothesis is that *all/at least one/none* of $\beta_j \neq$ _____ for the same j.

The test statistic is of the form:

$$\frac{\frac{a}{k}}{\frac{SSE}{n-k-1}}$$

What is the numeric value of a in the test? _____

What is the numeric value of the test statistic? _____

What is the critical value? _____

Which of the 4 options below is the correct conclusion from the test?

- Reject H0. There are significant differences in house prices across neighborhoods.
- Reject H0. There are no significant differences in house prices across neighborhoods.
- Don't reject H0. There are significant differences in house prices across neighborhoods.
- Don't reject H0. There are no significant differences in house prices across neighborhoods.

Answer:

As in Q11, let j be the neighborhood. The model is:

$$\mathbb{E}[Y|j] = \beta_0 + \beta_1 \mathbf{1}\{j = East\} + \beta_2 \mathbf{1}\{j = North\} + \beta_3 \mathbf{1}\{j = South\} + \beta_4 \mathbf{1}\{j = West\}$$

Under H_0, there are no differences in average house prices across neighborhoods, and so $\mathbb{E}[Y] = \beta_0$, i.e. the average price does not differ across neighborhoods. Under H_1, the average house price differs for at least one neighborhood and at least one β_j for $j = 1, 2, 3, 4$ is different from zero. We can test this with a standard F test.

- The null hypothesis is that *all* of $\beta_j = 0$ for j from 1 to 4.
- The alternative hypothesis is that *at least one* of $\beta_j \neq 0$ for the same j.
- The a represents the SSR. We can compute the value of this as follows

```
m0 <- lm(price ~ 1, data = df)
anova(m0, m4)
```

```
Analysis of Variance Table

Model 1: price ~ 1
Model 2: price ~ neighborhood
  Res.Df      RSS Df Sum of Sq      F    Pr(>F)
1    860 67528723
2    856 57327920  4  10200804 38.079 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The SSR is the value 10200804.

- The value of the test statistic is 38.079.
- The critical value is:

```
qf(0.95, 4, 856)
```

```
[1] 2.382332
```

- The test statistic exceeds the critical value so we reject H_0. There are significant differences in house prices across neighborhoods.

Model 5

Estimate a regression model with `price` as the dependent variable and the following 3 independent variables:

1. `area`
2. `bedrooms`
3. The interaction between `area` and `bedrooms`.

If you estimated the model correctly, your estimated intercept should equal 169.7672.

Answer:

```
m5 <- lm(price ~ area * bedrooms, data = df)
summary(m5)
```

```
Call:
lm(formula = price ~ area * bedrooms, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-597.99 -122.00    1.99   89.58  927.47

Coefficients:
              Estimate Std. Error t value   Pr(>|t|)
(Intercept)   169.7672    47.2150   3.596   0.000342 ***
area            6.3496     0.5752  11.040    < 2e-16 ***
bedrooms      -77.8622    15.4182  -5.050 0.00000054 ***
area:bedrooms   0.2520     0.1397   1.804   0.071554 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 185.4 on 857 degrees of freedom
Multiple R-squared:  0.5639,    Adjusted R-squared:  0.5624
F-statistic: 369.4 on 3 and 857 DF,  p-value: < 2.2e-16
```

Question 13

Which variables are individually statistically significant at the 5% level?

Answer: `area` and `bedrooms`. The interaction is not significant at the 5% level, but is significant at the 10% level.

Question 14

Obtain a 95% confidence interval for the coefficient in front of the variable `bedrooms`. Report the lower and upper bound.

Answer:

```
confint(m5)
```

```
                    2.5 %      97.5 %
(Intercept)     77.09666354 262.4378311
area             5.22066319   7.4784620
bedrooms      -108.12419338 -47.6002990
area:bedrooms   -0.02214487   0.5261141
```

Lower bound: -108.124 Upper bound: -47.600

Question 15

According to the model estimates, for a one-bedroom listing, by how much on average does the listing price increase (in €) when the area increases by 1m^2?

Answer: The model is:

$$\mathbb{E}[Y|area, bedrooms] = \beta_0 + \beta_1 area + \beta_2 bedrooms + \beta_3 area \times bedrooms$$

For one-bedroom listing:

$$\mathbb{E}[Y|area, bedrooms = 1] = \beta_0 + \beta_1 area + \beta_2 + \beta_3 area = \beta_0 + \beta_2 + (\beta_1 + \beta_3) area$$

Thus on average a one-unit increase in area increases the listing price by $\beta_1 + \beta_3$. We multiply this by 1000 to convert to euros.

We can calculate this with:

```
1000 * (coef(m5)[2] + coef(m5)[4])
```

```
    area
6601.547
```

Question 16

According to the model estimates, for a 100m^2 listing, by how much on average does the listing price change (in €) when it has one more bedroom (holding area fixed at 100m^2)?

Answer: The model is:

$$\mathbb{E}[Y|area, bedrooms] = \beta_0 + \beta_1 area + \beta_2 bedrooms + \beta_3 area \times bedrooms$$

For listings with 100m^2:

$$\mathbb{E}[Y|area = 100, bedrooms] = \beta_0 + 100\beta_1 + \beta_2 bedrooms + \beta_3 \times 100 \times bedrooms$$

Thus the coefficient in front of `bedrooms` is $\beta_2 + 100\beta_3$. We multiply this by 1000 to convert to euros.

```
1000 * (coef(m5)[3] + 100 * coef(m5)[4])
```

```
 bedrooms
-52663.78
```